

Marginally Useful: Formalizing the Information Gap in Conformal Prediction

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Abstract

Conformal prediction supplies a finite-sample, distribution-free guarantee: a prediction set that covers the outcome with at least a stated probability, averaged over inputs. The guarantee is genuine, and it is routinely misread as evidence that the forecast underneath is good. We separate the two. For a fixed location predictor whose residuals are re-leveled by a single shared shape—the setting in which conformal prediction is used to emit a predictive distribution—we show that the expected logarithmic-score shortfall relative to the best possible forecast equals one information-theoretic quantity: the mutual information between the residual and the input. Conformalization re-levels coverage but cannot reduce this quantity, because it is a property of the predictor’s shape class rather than of calibration; no recalibration that ignores the input can touch it. Three familiar cautions follow as consequences rather than slogans: marginal coverage is not conditional, validity is insensitive to sharpness, and the guarantee requires exchangeability. We give a two-coordinate diagnostic that keeps coverage and forecast quality visibly distinct, and we delineate the tasks for which coverage is itself the right objective, where conformal prediction is not merely sound but ideal.

Keywords: Conformal prediction; proper scoring rules; calibration and sharpness; mutual information; predictive distributions.

1 Introduction

A practitioner maintains a probabilistic forecaster and is told, repeatedly, that conformal prediction is the principled, distribution-free way to “add uncertainty.” They bolt it on. The number they actually report—a proper score such as the predictive log-likelihood—does not improve. The natural diagnosis is a tuning failure. The real one is a *category error*: conformal prediction is a method for *certifying the coverage of a set*, and it is being asked to *estimate a distribution*. These are different kinds of object, graded by different instruments, and no amount of tuning turns one into the other. None of this is a criticism of the method. It is a guide to where the method belongs.

The mathematics is not in question. Given exchangeable data, a fitted predictor, and a nonconformity score, split conformal prediction returns a set $C_\alpha(x)$ with

$$\mathbb{P}(Y_{n+1} \in C_\alpha(X_{n+1})) \geq 1 - \alpha, \quad (1)$$

finite-sample and with no assumption on the data distribution (Vovk et al., 2005; Lei et al., 2018; Angelopoulos and Bates, 2023). The trouble is entirely in the reading of (1). The probability averages over X_{n+1} as well as Y_{n+1} : it is a statement about the *average* future input, not about the one in front of you. It is *marginal* coverage, and as the foundational analyses warned from the start, “a good estimator must satisfy something more than marginal coverage” (Lei and Wasserman, 2014).

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Contribution. This paper makes that warning exact. Our one new result (Section 6) is an identity. Fix a location predictor and let conformal prediction re-level its residuals by a single shared shape—the conformal predictive system of Vovk et al. (2019), which is precisely the mode in which conformal prediction outputs a full predictive distribution. Then the expected log-score regret of that distribution relative to the conditional oracle is *exactly* the mutual information $I(R; X)$ between the residual R and the input X (Proposition 3). We call this the residual-information gap. It is the quantity conformalization cannot touch: re-leveling that ignores X can change the shape but not the dependence, so it leaves $I(R; X)$ invariant. Reducing the gap requires modeling the conditional residual—heteroscedastic scale, quantiles, or shape—which is exactly the activity the conformal step does not perform. The identity admits five equivalent readings (Section 6): a false-pooling cost, an average log Bayes factor, the conditional non-uniformity of conformal ranks, an information projection, and a Kelly betting rent.

What is, and is not, new. We are explicit about provenance. The impossibility of distribution-free conditional coverage is due to Lei and Wasserman (2014) and Foygel Barber et al. (2021); the re-leveling object is the conformal predictive system of Vovk et al. (2019); the decomposition of a proper score into calibration and sharpness is standard (Gneiting et al., 2007; Gneiting and Raftery, 2007). The contribution is to combine these into one exact identity along the score axis, to read it five ways, and to draw from it a two-coordinate diagnostic (Section 7) on which conformalization moves a report horizontally—toward exact coverage—and never vertically, toward a better score. Our quarrel is with a conflation, not with conformal prediction: when coverage is itself the deliverable (Section 9), the method is not merely sound but often ideal.

Structure. Section 2 places the result in the literature. Section 3 fixes notation. Section 4 restates, with attribution, three known limits of the marginal guarantee. Section 5 shows coverage and the log-score are orthogonal for a residual score; Section 6 proves the residual-information gap; Section 7 draws the coverage–score plane. Section 8 treats exchangeability and time series, Section 9 the case where coverage is the objective, and Section 10 the relation to the modern conformal literature.

2 Related work

The conflation we address. Applied and survey writing routinely calls the marginal-coverage guarantee “well-calibrated” or “reliable” uncertainty quantification (Bellotti and Zhao, 2025), and tutorials present conformal prediction as the way to “add uncertainty” (tds) or to “predict full probability distributions” (Manokhin). The methodological literature is more careful, separating *validity* from *efficiency*; our quarrel is with the conflation, not with the method. A recent critique strand makes related points independently: that valid sets can be uninformative (Recht, 2024), that interval length can be gamed while marginal coverage is held fixed (Min et al., 2026), and that the guarantee is misread in applied domains (Mehrtens et al., 2025). We add an exact identity for the log-score regret, which these treat operationally rather than in closed form.

Foundations and the marginal guarantee. Split conformal prediction and its finite-sample marginal validity are due to Vovk et al. (2005) and Lei et al. (2018); see Angelopoulos and Bates (2023) and Angelopoulos et al. (2024a) for modern treatments. The conformal predictive system (Vovk et al., 2019) is the construction that turns a conformal predictor into a full predictive distribution by re-leveling the marginal residual law; it is the object our regret identity grades.

The limits of marginal coverage. That marginal coverage is not conditional, and that distribution-free conditional coverage is unattainable at finite length, is due to Lei and Wasserman (2014) and

sharpened by Foygel Barber et al. (2021); Vovk (2012) studies conditional validity for inductive conformal predictors. Approaches that approach the conditional goalpost under extra structure include Gibbs et al. (2025) and Laplante (2026), and diagnostics that quantify the conditional gap include Braun et al. (2025). These condition on X or assume structure; none closes the gap distribution-free, consistent with the impossibility results.

Proper scores, calibration, and recalibration. The calibration/sharpness decomposition and the maximize-sharpness-subject-to-calibration program are due to Gneiting et al. (2007); Gneiting and Raftery (2007), with antecedents in the prequential view of Dawid (1984). Recalibration methods (Platt, 1999; Zadrozny and Elkan, 2002; Guo et al., 2017; Kuleshov et al., 2018) adjust a forecast’s probabilities; we use the log-score throughout and relate what X -blind recalibration can and cannot do to the gap.

Information-theoretic and conditional conformal. Closest to us is Correia et al. (2024), who connect conformal set size to information-theoretic uncertainty via bounds on expected set size in terms of the conditional entropy $H(Y | X)$. That work grades the *coverage/efficiency* axis; we grade the *score* axis, and our identity is for the log-score regret of the predictive distribution rather than for set size. The two are complementary, and we make the connection precise in Section 7. Conditional and shape-adaptive conformal constructions that leave the single-shape class—conformalized quantile regression (Romano et al., 2019), Mondrian/binmed conformal predictive systems (Boström et al., 2021; Toccaceli, 2026), and conformal training (Stutz et al., 2022)—are discussed in Section 10, where we show they confirm the paper’s mechanism: they buy sharpness by conditioning on X or optimizing a proper objective, which is exactly what the gap says is required.

3 Setup: split conformal and the marginal guarantee

Let $(X_i, Y_i)_{i=1}^{n+1}$ be exchangeable. Fix a point predictor $\hat{\mu}$ trained on a disjoint set, and a *nonconformity score* $s(x, y)$; the canonical choice in regression is the absolute residual $s(x, y) = |y - \hat{\mu}(x)|$. Compute calibration scores $s_i = s(X_i, Y_i)$ for $i = 1, \dots, n$ and let

$$\hat{q} = s_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil, \quad (2)$$

the k -th order statistic (with $\hat{q} = +\infty$ when $k > n$). The split-conformal set is

$$C_\alpha(x) = \{y : s(x, y) \leq \hat{q}\} = [\hat{\mu}(x) - \hat{q}, \hat{\mu}(x) + \hat{q}] \quad (3)$$

for the absolute-residual score. Exchangeability of the augmented scores makes the rank of s_{n+1} uniform, which yields (1) (Vovk et al., 2005; Lei et al., 2018). The construction is simple and the guarantee is real, and one readily verifies that empirical coverage tracks $1 - \alpha$ as the sample size, noise level, and α are varied. None of this is in dispute. The question is what it is taken to mean.

4 Three known limits of the marginal guarantee

None of the three facts below is new. We restate them, with attribution, to fix notation and to mark where the contribution of Section 6 departs from what is already established.

(i) *Marginal is not conditional, and conditional is impossible for free.* One would prefer *conditional* coverage, $\mathbb{P}(Y \in C(x) | X = x) \geq 1 - \alpha$ for (almost) every x . Distribution-free, this cannot be bought. Lei and Wasserman (2014) show (their Lemma 1) that any band with non-trivial finite-sample

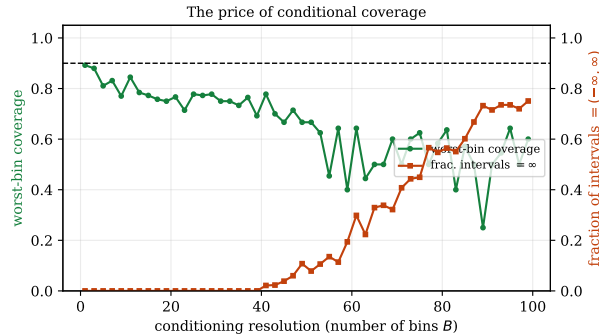


Figure 1: The price of conditional coverage (Lei and Wasserman, 2014). As the conditioning resolution B grows, per-cell calibration sets shrink and the only valid cell interval becomes $(-\infty, \infty)$; uniform (worst-bin) coverage is bought only as the fraction of infinite-length intervals rises.

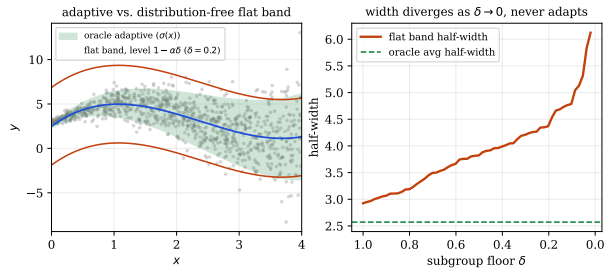


Figure 2: Subgroup coverage buys only a wider band (Foygel Barber et al., 2021). The distribution-free protection of all δ -subgroups is a flat, non-adaptive band at level $1 - \alpha\delta$ whose width diverges as $\delta \rightarrow 0$; the adaptive oracle that uses $\sigma(x)$ is far tighter but requires the distributional assumption.

conditional validity has *infinite* expected length at almost every point of a continuous distribution. Foygel Barber et al. (2021) restate and sharpen this: a conditionally valid C_n has $\mathbb{E}\text{leb}(C_n(x)) = \infty$ at almost all non-atomic x (their Prop. 2.2), and even the relaxed demand of coverage on every subgroup of probability $\geq \delta$ “is impossible to attain beyond the trivial solution” of inflating the marginal level to $1 - \alpha\delta$ (their Thm. 3.1). These are not loose bounds to be tightened; they are the reason the gap cannot be closed for free, and both surface as concrete finite-sample facts. The first is illustrated in Figure 1: localizing split conformal to recover per- x coverage shrinks each cell’s calibration count n_b until $\lceil (n_b + 1)(1 - \alpha) \rceil > n_b$, at which point (2) gives $\hat{q} = +\infty$ and the only valid cell interval is $(-\infty, \infty)$; uniform coverage is recovered only as the fraction of infinite-length intervals grows. The second is Figure 2: the distribution-free protection of all δ -subgroups is exactly the flat band at level $1 - \alpha\delta$, whose width diverges as $\delta \rightarrow 0$ while never adapting to x , at a multiplicative width cost over an assumption-driven adaptive oracle.

So the guarantee you can have distribution-free is the marginal one, and a marginally valid band “tends to overestimate the set when x is in the high density area and to underestimate for low density x ” (Lei and Wasserman, 2014): it over-covers the easy inputs and under-covers the hard ones, silent by construction about the cases the model was built to get right. Figure 3 exhibits exactly this on heteroscedastic data: 90% overall, near-100% where the noise is small, well below target where it is large.

The conformal literature already separates *validity* from *efficiency*. The point here is only that the marginal-coverage certificate does not measure distributional quality.

(ii) *Validity is trivially satisfiable.* A predictor that returns the whole outcome space with probability $1 - \alpha$ and the empty set otherwise satisfies (1) exactly and conveys nothing. More practically, the *same* coverage attaches to an excellent $\hat{\mu}$ and to a deliberately terrible one; the latter merely yields a wider $\hat{q}$ in (2). Marginal validity is therefore not evidence of a good method; it certifies the fence, not the cattle. The next section makes this precise.

(iii) *Without exchangeability, even the marginal guarantee fails.* Equation (1) rests on exchangeability of the augmented sample. Under covariate shift, label shift, or temporal dependence it does not hold, and the patches that restore something do so by weakening what is guaranteed (Section 8).

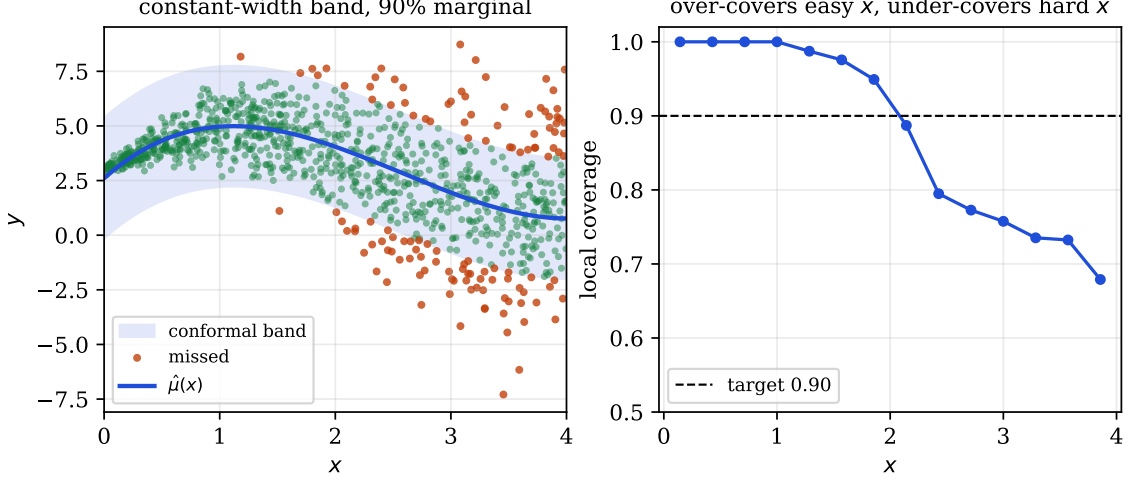


Figure 3: Marginal is not conditional. A constant-width split-conformal band (left) attains 90% *marginal* coverage on heteroscedastic data, but its *local* coverage (right) runs near 100% where the noise is small and well below target where it is large: it over-covers the easy inputs and under-covers the hard ones.

5 Coverage and score are orthogonal

We first make “coverage is not a measure of forecast quality” a precise, constructive fact. Separate the two things a forecaster reports: a point predictor $\hat{\mu}(x)$ and a full predictive density $f(\cdot | x)$. The log-score grades f , that is, its spread and shape, its sharpness. The canonical residual score $s(x, y) = |y - \hat{\mu}(x)|$ uses only $\hat{\mu}$, the *location* of the forecast. It never sees the spread or shape of f . So the conformal set, and its coverage, are invariant to exactly the part of f that the log-score rewards. (This is specific to such scores; it does not apply to density- or quantile-based scores, which read f directly.)

Proposition 1 (Residual-score coverage does not constrain the log-score). *The split-conformal set (3) is a measurable function of the nonconformity scores $\{s_i\}$ and of $s(x, \cdot)$; it is not a functional of the predictive density f unless f is explicitly used in the score. Consequently two forecasters with the same location $\hat{\mu}$ (hence, for the residual score, the same score function and calibration scores) but different predictive densities f have identical conformal sets, and identical marginal coverage at every level α , while their expected log-scores $\mathbb{E}[\log f(Y | X)]$ may differ by an arbitrary amount.*

Proof. The first claim is immediate from (2)–(3): $\hat{q}$ is an order statistic of $\{s_i\}$ and $C_\alpha(x) = \{y : s(x, y) \leq \hat{q}\}$, so two forecasters agreeing on s and on $\{s_i\}$ produce the same set for every x, α , and (1) concerns only that set.

For the gap, take X degenerate (a point mass at an arbitrary x_0 , i.e. a single repeated input, so there is nothing to vary across) and $Y \sim N(0, 1)$. Both forecasters share the location $\hat{\mu} \equiv 0$, so both use the absolute-residual score $s(x, y) = |y|$, which depends on $\hat{\mu}$ but not on the spread of the predictive density. They differ only in that claimed density, $f_\sigma = N(0, \sigma^2)$: for any σ the scores are $|Y_i|$, independent of σ , so the conformal set is the fixed interval $[-\hat{q}, \hat{q}]$ with $\hat{q}$ the empirical $(1 - \alpha)$ -quantile of $|Y|$. Yet

$$\mathbb{E}[\log f_\sigma(Y)] = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{\mathbb{E}[Y^2]}{2\sigma^2} = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \xrightarrow{\sigma \rightarrow 0^+} -\infty,$$

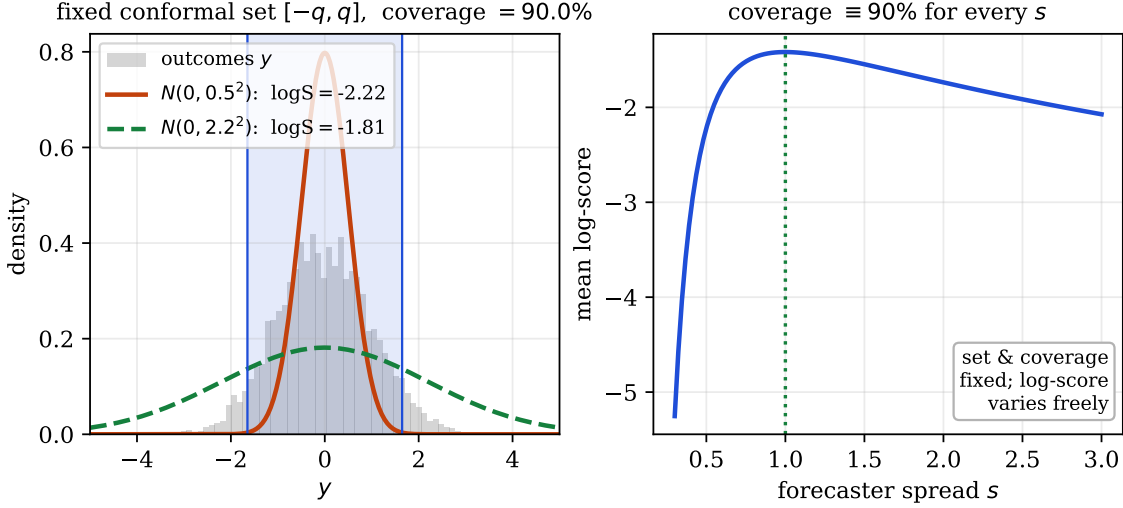


Figure 4: Coverage does not constrain the log-score (Proposition 1). With a residual score the conformal set $[-q, q]$ is fixed by the outcomes alone (left), so two predictive densities with very different log-scores share the *same* set and the same 90% coverage. Sweeping the forecaster’s spread s (right) moves the log-score freely while coverage stays pinned at 90%.

and likewise $\rightarrow -\infty$ as $\sigma \rightarrow \infty$. Hence for any $M > 0$ there are σ_1, σ_2 with identical conformal sets (identical marginal coverage at every α) yet log-scores differing by more than M . $\square$

Corollary 1. *For a residual-score conformal predictor, marginal validity certifies an order statistic of residual magnitudes. It does not identify, bound, or rank the scale, shape, or sharpness of any predictive density that was not used to construct the score; in particular, no inference about forecast quality (as graded by a strictly proper score) can be drawn from coverage alone.*

Remark 1 (The degenerate X is only for cleanliness). The point-mass X in the proof makes the unbounded gap transparent, but it is not needed. For any non-degenerate X , take two forecasters sharing the same location $\hat{\mu}$ (hence, for the residual score, the same score function and calibration scores, so identical sets and coverage at every α) but predictive spreads $\sigma_1(x), \sigma_2(x)$; their expected log-scores differ by $\mathbb{E}_X[\log(\sigma_2/\sigma_1)] + \frac{1}{2}\mathbb{E}_X[(\sigma_2^{-2} - \sigma_1^{-2}) \text{Var}(Y | X)]$, which is unbounded over choices of the σ_i . Coverage is blind to all of it.

This is the formal version of a point several authors have made informally: that conformal validity “almost invites you to use garbage prediction functions” (Recht, 2024). Figure 4 shows it: the claimed density’s spread slides freely while the conformal interval and its 90% coverage do not move, because the spread never entered (2).

6 The residual-information gap

Forecast quality decomposes into *calibration* and *sharpness*, and the discipline’s stated goal is to maximize sharpness subject to calibration, assessed by a proper scoring rule (Gneiting et al., 2007; Gneiting and Raftery, 2007). Within the fixed-location, single-shape residual class, signed CPS is log-score optimal; the loss is the class restriction, not the conformal calibration step.

Conformal predictive systems. A conformal predictor can be made to output a distribution rather than a set, the *conformal predictive system* (CPS) of Vovk et al. (2019), built from *signed* residual scores. It is the mode in which conformal prediction estimates a full distribution, so it is the form the criticism must address. We must, however, distinguish it from the more common practice of reading the nested symmetric intervals of ordinary *absolute*-residual split conformal as if they were a distribution; the two re-level different objects, and the distinction matters for the gap below.

Proposition 2 (Re-leveling, two cases). *Fix a location predictor $\hat{\mu}$ with residual $R = Y - \hat{\mu}(X)$ of marginal density g and CDF G . In the large-calibration limit (under exchangeability):¹*

- (A) *Signed-residual CPS. The CPS predictive distribution is $\Pi_x(y) = G(y - \hat{\mu}(x))$ (the calibration ECDF $\hat{G}_n \rightarrow G$ by Glivenko–Cantelli), with density $\pi_x(y) = g(y - \hat{\mu}(x))$: the base location forecast re-leveled by the marginal signed-residual law.*
- (B) *Absolute-residual intervals. Ordinary split conformal at level α returns the symmetric interval $\hat{\mu}(x) \pm \hat{q}_\alpha$, with $\hat{q}_\alpha$ the $(1 - \alpha)$ -quantile of $|R|$. Read as a predictive distribution (by sweeping α) it is symmetric about $\hat{\mu}(x)$ with $|\cdot|$ distributed as $|R|$; its density is the symmetrized marginal residual law $h_{\text{sym}}(z) = \frac{1}{2}(g(z) + g(-z))$. This is not an additional conformal guarantee; it is the density implicitly obtained if the nested absolute-residual intervals are read as central credible intervals.*

Proof sketch. (A) The CPS at level α returns endpoints equal to the signed-residual order statistics; sweeping α traces $\hat{\mu}(x)$ plus the residual quantile function, i.e. $\hat{G}_n^{-1}$ shifted by $\hat{\mu}(x)$, so $\Pi_x(y) = \hat{G}_n(y - \hat{\mu}(x))$; take limits (Vovk et al., 2019). (B) The intervals depend on the data only through $|R|$ and are symmetric about $\hat{\mu}(x)$; the unique symmetric density whose absolute value has the law of $|R|$ (with signed marginal g) is $h_{\text{sym}}(z) = \frac{1}{2}(g(z) + g(-z))$. $\square$

In both cases the estimation was done by the base model and the residual sample; conformal supplies only the leveling. We now quantify the cost. Write the true conditional residual density as $r(\cdot | x)$, so the oracle density is $q^*(y | x) = r(y - \hat{\mu}(x) | x)$, and let $\bar{r} = \mathbb{E}_X r(\cdot | X) = g$ be the marginal residual density.

Proposition 3 (Residual-information gap). *Fix the location predictor $\hat{\mu}$ and write $I(R; X) = \mathbb{E}_X \text{KL}(r(\cdot | X) \| \bar{r}) \geq 0$, the mutual information between the residual and the input. In the large-calibration limit of Proposition 2 (so that the signed-CPS shape is $\bar{r}$ and the absolute-residual shape is h_{sym}), and assuming the relevant densities and information quantities are finite, the expected log-score regret relative to the oracle q^* is*

$$(A) \text{ signed CPS: } \mathbb{E}[\log q^*(Y | X)] - \mathbb{E}[\log \pi_X(Y)] = I(R; X), \quad (4)$$

$$(B) \text{ absolute intervals: } \mathbb{E}[\log q^*(Y | X)] - \mathbb{E}[\log h_{\text{sym}}(R)] = I(R; X) + \text{KL}(\bar{r} \| h_{\text{sym}}). \quad (5)$$

Both are ≥ 0 . The term $I(R; X)$ vanishes iff $R \perp X$; the extra term $\text{KL}(\bar{r} \| h_{\text{sym}})$ vanishes iff the marginal residual law is symmetric. Among all single-shape forecasters $y \mapsto h(y - \hat{\mu}(x))$, $\mathbb{E}[\log h(R)]$ is maximized at $h = \bar{r}$; the signed CPS attains this optimum. Hence no recalibration that ignores X can reduce $I(R; X)$: it can at best replace the shape by $\bar{r}$, removing the skewness term in (5) and reducing case (B) to case (A). Reducing $I(R; X)$ itself requires conditioning on X .

¹The log-score comparison is for the limiting residual density, or equivalently a smoothed CPS density: a finite empirical CPS is a valid predictive distribution but not generally a Lebesgue density, so the density/log-score identity is an asymptotic (or smoothed-density) statement. The finite-sample guarantee is coverage.

Proof. Write H for the residual law with density h , and let $\mathcal{R}(h)$ be the expected log-score regret of the single-shape forecaster $y \mapsto h(y - \hat{\mu}(x))$ relative to the oracle. The oracle scores Y by $\log q^*(Y|X) = \log r(R|X)$ and this forecaster by $\log h(R)$; the joint $P_{X,R}$ has density $p(x)r(\rho|x)$ and the product reference $P_X \otimes H$ has density $p(x)h(\rho)$, so

$$\mathcal{R}(h) = \mathbb{E} \left[\log \frac{r(R|X)}{h(R)} \right] = \text{KL}(P_{X,R} \| P_X \otimes H) = I(R; X) + \text{KL}(\bar{r} \| h), \quad (6)$$

the last step the chain rule for relative entropy along a product reference (the R -marginal of $P_{X,R}$ is $\bar{r}$, and $I(R; X) = \text{KL}(P_{X,R} \| P_X P_R)$). Taking $h = \bar{r}$ kills the second term and gives (4); taking $h = h_{\text{sym}}$ gives (5). The second term is ≥ 0 and is minimized at $h = \bar{r}$ (Gibbs), so no X -blind shape improves on $I(R; X)$, and $I(R; X) = \text{KL}(P_{X,R} \| P_X P_R)$ is reduced only by conditioning on X . $\square$

Remark 2 (What this says, and does not say). The right-hand side of (4) is exactly the information about the residual distribution that remains in X after the location predictor is fixed. Calling this the residual-information gap makes the content of “blind to conditional residual shape” precise. Three caveats. (1) *Conformal is not dominated within its class.* The signed CPS is log-score optimal among single-shape location forecasters. What it pays for is the class restriction, not the calibration step. The absolute-residual interval system is the one that pays the extra $\text{KL}(\bar{r} \| h_{\text{sym}})$, for discarding sign information; since $h_{\text{sym}} \geq \frac{1}{2}\bar{r}$ this penalty is at most $\log 2$, one bit, so the structural loss is $I(R; X)$. (2) *What recalibration can and cannot do.* A recalibration that ignores X (e.g. marginal PIT recalibration) can move the shape toward $\bar{r}$ and thereby erase the skewness term, i.e. turn case (B) into case (A). It cannot touch $I(R; X)$. Reducing $I(R; X)$ requires modeling conditional residual shape: heteroscedastic models, conditional recalibration (Kuleshov et al., 2018), or conformalized quantile regression (Romano et al., 2019), which is the conformal world’s way of leaving the single-shape class. (3) *The distribution-free guarantee is real.* Recalibration offers calibration only in expectation or asymptotically and carries no coverage certificate; conformal’s marginal guarantee is finite-sample and assumption-light. When the certificate itself is the deliverable, that is decisive (Section 9).

Remark 3 (Five readings of the gap). The identity $\mathcal{R}(\bar{r}) = I(R; X) = \text{KL}(P_{X,R} \| P_X P_R)$ admits several readings, each a different angle on the same number. (i) *A false-pooling cost.* A single-shape system asserts that, after $\hat{\mu}$, one residual law fits everyone, i.e. $R \perp X$; the gap is the relative entropy from the true residual experiment to the nearest such independence model, the log-score price of that assumption. (ii) *An average log Bayes factor.* Since $I(R; X) = \mathbb{E} \log \frac{r(R|X)}{\bar{r}(R)}$, each observation contributes the log-likelihood ratio of its x -specific residual law against the pooled one, and the gap is the oracle’s expected per-sample advantage. (iii) *Conditional non-uniformity of conformal ranks.* Let $U = G(R)$ be the rank of the residual under the pooled law, the PIT used by signed CPS (assuming G is continuous; use the randomized PIT otherwise). Marginally U is uniform, which is the conformal achievement; but U is an invertible transform of R , so $I(R; X) = \mathbb{E}_X \text{KL}(P_{U|X} \| \text{Unif}[0, 1])$ is the *conditional* non-uniformity of those ranks. Easy inputs concentrate U near $\frac{1}{2}$, hard inputs push it into the tails, and the single pooled shape cannot tell them apart (Figure 5). (iv) *A projection.* Equation (6) is the information projection of $P_{X,R}$ onto the single-shape product family $\{P_X \otimes H\}$. The optimizer is $H = P_R$, that is $P_X \otimes P_R$, and the projection loss is $\text{KL}(P_{X,R} \| P_X P_R) = I(R; X)$. No X -blind step can cross that loss, because the missing information is dependence, not marginal shape. (v) *A betting rent.* By the log-optimal (Kelly) growth theorem, a gambler who knows a distribution P and bets against a fair book priced by Q compounds wealth at expected log-rate $\text{KL}(P \| Q)$ per round (Kelly, 1956; Cover and Thomas, 2006). Reading $\bar{r}$ as the book of a single-shape conformal forecaster and $r(\cdot|X)$ as the oracle’s

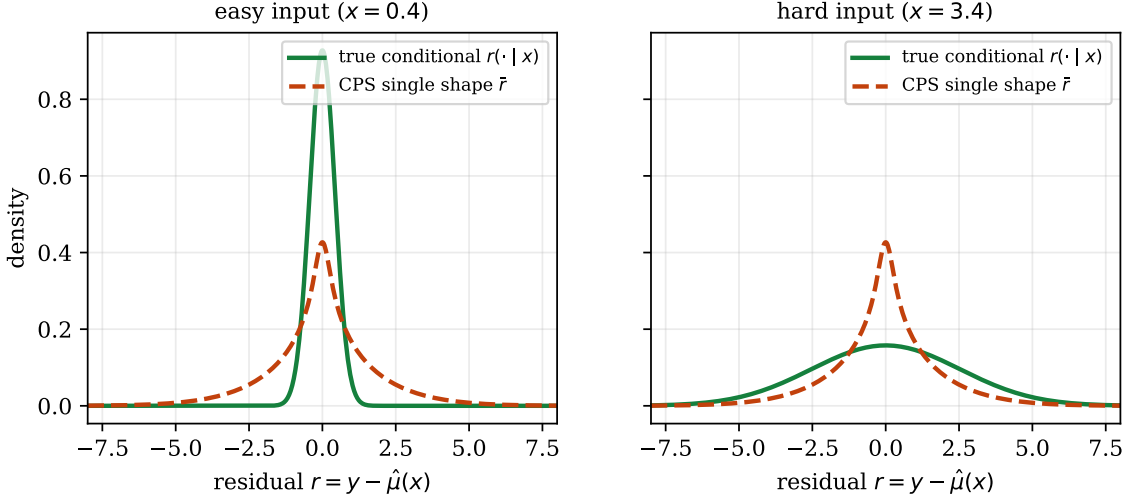


Figure 5: The residual-information gap (Propositions 2–3). A single-shape conformal predictive system uses the marginal residual law $\bar{r}$ for every input; the oracle uses the true conditional $r(\cdot | x)$, narrow for easy inputs (left) and wide for hard ones (right). The expected log-score regret is exactly the mutual information $I(R; X)$, which no X -ignoring recalibration can reduce.

knowledge, the oracle’s expected per-round rent is $\mathbb{E}_X \text{KL}(r(\cdot | X) \| \bar{r}) = I(R; X)$, the same quantity as in (ii) measured in log-wealth. The conformal step fixes the level of the book (its normalization, hence coverage), not the rent: a re-leveling that ignores X leaves the divergence, and so the growth rate, unchanged.

Remark 4 (The gap is not an artifact of a poor location fit). The quantity $I(R; X)$ is defined for a *fixed* $\hat{\mu}$, so a natural worry is that it merely measures how badly $\hat{\mu}$ fits. It does not. Even the Bayes-optimal location $\hat{\mu}(x) = \mathbb{E}[Y | X = x]$, which zeroes the conditional mean of R , leaves a residual whose *conditional law* still varies with x under heteroscedasticity or changing shape, so $I(R; X) > 0$ in general; the gap measures conditional residual *scale and shape* information, which is orthogonal to getting the location right. Improving $\hat{\mu}$ can raise or lower $I(R; X)$, but the lever the gap identifies is conditioning the residual *shape* on X , not refining the point forecast.

The pieces this assembles are individually known (Section 2); the addition is a single exact identity along the score axis. For the single-shape system the log-score regret to the oracle equals the conditional residual information $I(R; X)$, with an additional symmetrisation term for absolute-residual intervals, so the quantity conformalization cannot touch has a name and a closed form. We have not seen the regret stated this way.

Figure 5 draws the gap: under heteroscedasticity the true conditional residual law $r(\cdot | x)$ varies with x , while the single-shape system uses the marginal $\bar{r}$ everywhere, and the regret is exactly the information $I(R; X)$ that separates them. The special case $I(R; X) = 0$ ($\hat{\mu}$ correct and r a fixed Gaussian with the wrong scale) is the one where variance recalibration fit to the log-score recovers the shape and improves the score, while conformalization re-levels to exact coverage and returns a set.

7 The coverage–score plane

The preceding results suggest a single diagnostic, provided we are careful about what is being plotted. Define a *reporting system* as a set-valued report $C_\alpha(x)$ and, optionally, a density-valued report $f(\cdot | x)$. Its coordinates are

$$(C, S), \quad C = |\mathbb{P}(Y \in C_\alpha(X)) - (1 - \alpha)|, \quad S = \mathbb{E}[\log f(Y | X)], \quad (7)$$

the marginal-coverage error of the *set* and the proper-score performance of the *density*. The decomposition of forecast quality into calibration and sharpness (Gneiting et al., 2007) is, in this plane, the statement that the two coordinates are distinct, and our results read geometrically:

- Residual split conformal is a horizontal projection. It maps a report $(C_\alpha^{\text{base}}, f)$ to $(C_\alpha^{\text{conf}}, f)$: it changes the set so that $C \rightarrow 0$ while leaving the density f (and hence S) untouched (Proposition 1). It moves a point left, never up.
- Conformal predictive systems are different: they supply their own density-valued report, so they do have a vertical coordinate, but that coordinate is governed by the single-shape restriction of Proposition 3, capped at the oracle minus $I(R; X)$.
- Recalibration and better modeling move vertically. Fitting shape to a proper score raises S ; conditioning on x is the only way to raise it past the ceiling set by the residual-information gap (4).

A system reported by its coverage alone is reported by one coordinate of a two-coordinate quality. The plane is a presentation device (adjacent to reliability and sharpness diagrams (Gneiting et al., 2007)), but the “conformal = move left, never up” reading makes the category distinction hard to miss. This also sharpens the connection to Correia et al. (2024): their set-size/entropy bound lives on the *efficiency* coordinate, the information-bearing one, while the marginal-coverage coordinate, by Proposition 1, carries none. Figure 6 is the schematic. Conformalization is the horizontal move (left, toward exact coverage, never up); X -blind recalibration is the vertical move, but only up to the single-shape ceiling; and the residual-information gap $I(R; X)$ that separates that ceiling from the oracle is crossed only by conditioning on X .

8 Exchangeability and the time-series case

Drop exchangeability and (1) is no longer guaranteed. The literature’s repairs are ingenious, and uniform in one respect: they survive by redefining what is guaranteed.

- Weighted conformal prediction (Tibshirani et al., 2019) restores marginal coverage under covariate shift when the likelihood ratio is known; with estimated weights coverage becomes approximate and the effective sample size drops.
- Adaptive Conformal Inference (Gibbs and Candès, 2021) updates the level online and “puts no constraints on the data generating distribution,” but guarantees a *long-run time-average* miscoverage frequency, $\frac{1}{T} \sum_t \text{err}_t \rightarrow \alpha$, not a finite-sample statement about $\mathbb{P}(Y \in C(X))$, and certainly not conditional coverage.
- EnbPI (Xu and Xie, 2021) drops exchangeability for approximate marginal coverage under strong-mixing/stationarity surrogates; later online methods (Zaffran et al., 2022; Gibbs and

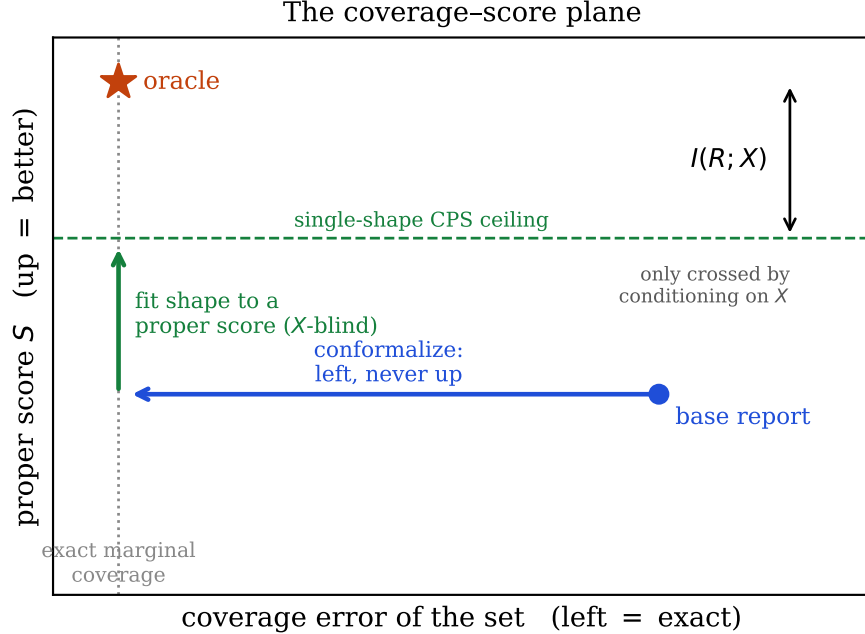


Figure 6: The coverage–score plane (schematic). The two coordinates are distinct: conformalizing a residual-score report moves it horizontally toward exact marginal coverage without changing its proper score (Proposition 1); fitting the residual shape to a proper score moves it up, but an X -blind shape can rise only to the single-shape ceiling (Proposition 3); the remaining residual-information gap $I(R; X)$ to the oracle is closed only by conditioning on X .

Candès, 2024) retreat to regret bounds over local intervals. The general beyond-exchangeability analysis of Barber et al. (2023) bounds the coverage gap by a total-variation distance that is exactly what nonstationarity inflates.

Three distinct guarantees now travel under the word “conformal”: (i) *split conformal under exchangeability* gives finite-sample marginal coverage; (ii) *adaptive conformal inference* gives long-run, time-average coverage control under distribution shift; (iii) *online conformal under arbitrary shifts* gives regret-style or local-interval performance rather than the original exchangeable guarantee. These are progressively weaker and more diffuse objects; none is the sharp predictive distribution *now*, for *this* step, that a proper score grades. This takes nothing from their utility: long-run, time-average frequency control under drift is a genuine and hard-won engineering guarantee, often the most one can ask for without a model of the shift. It is simply a weaker, differently posed object than conditional coverage *now*. The exchangeability required by guarantee (i) is typically unavailable in deployed time-series settings, though blocking or learned-transform constructions can sometimes recover an exchangeability-like assumption.

One illustration makes the consequence concrete (Figure 7). On a nonstationary series, fixed split conformal collapses far below its nominal coverage once the volatility regime shifts, because exchangeability has failed; the online adaptive variants recover a long-run coverage *frequency* by re-leveling to recently realized residuals. This is consistent with the gap of Section 6: where adaptivity helps it does so by *local conditional modeling* of the error scale, with the conformal step supplying the coverage leveling on top, never the sharpness. A simple probabilistic model that estimates the conditional spread directly—for instance a streaming exponentially-weighted

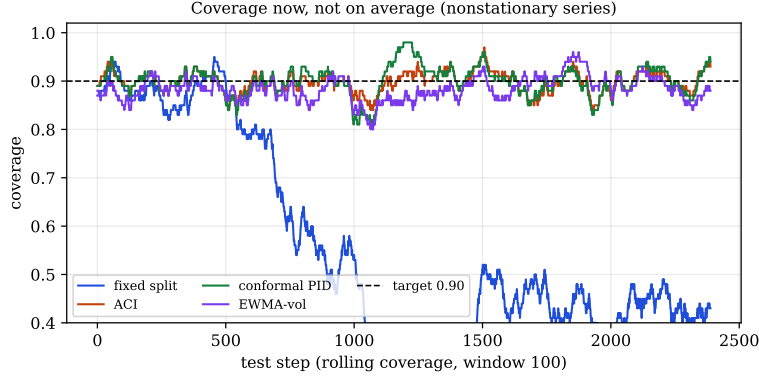


Figure 7: Coverage now, not on average. Rolling coverage on a nonstationary series: fixed split conformal collapses once the volatility regime arrives, while adaptive conformal and a probabilistic volatility model track the target. Even the recovering methods oscillate around the target rather than holding it pointwise.

volatility forecaster (Cotton, 2021)—tracks the target and returns a full predictive distribution, and conformalizing it leaves any proper score essentially unchanged, which is the experience the introduction describes.

9 When coverage *is* the objective

The argument is skeptical, not dismissive, and it comes with a precise converse. The litmus is one question:

Is your loss a function of whether the truth lands in a region, or of where it lands?

If the former, coverage is the native object and conformal prediction is sound and often ideal:

- Selective prediction and risk control. “Return a small set guaranteed to contain the label 95% of the time; a human checks it.” The deliverable is a set; risk-controlling prediction sets and conformal risk control formalize this (Angelopoulos et al., 2021; Bates et al., 2021; Angelopoulos et al., 2024b).
- Retrieval and shortlisting, where the product is a candidate set and coverage is recall.
- Anomaly and novelty detection via conformal p -values, conformal prediction’s most natural home, because here it is not a forecaster but a distribution-free hypothesis test and coverage is Type-I error control (Vovk et al., 2005).
- Compliance and long-run frequency control, where a certificate of the form “contained 95% of the time on average” *is* the product, and the distribution-free, finite-sample nature of (1) is exactly the value added.

Two caveats temper even these. If you trust a calibrated posterior you can form a sharper set yourself (the smallest region of mass $\geq 1 - \alpha$) and recover the expectations a bare set discards; conformal’s marginal value is then purely the distribution-free guarantee, useful when you distrust the density and exchangeability holds. And even when coverage is the goal you usually want it conditionally, which returns us to the impossibility of Section 4.

10 Scope, and the constructive escapes

Our strong claims target the object that is actually over-sold: *post-hoc, marginal, split conformal prediction with a residual score*. A large and active literature builds conformal methods that do more; the conditional constructions below (CQR, Mondrian / binned CPS) are by now standard practice, not niche alternatives, and we do not question their effectiveness. How they relate still matters, because in every case they confirm the paper’s mechanism rather than contradict it: they buy sharpness or conditional coverage by *conditioning on X or optimizing a proper objective*, which is exactly what Proposition 3 says is required — that is, by modeling, rather than by anything the conformal step itself supplies.

Conformal predictive systems that condition on X . Mondrian / binned conformal predictive systems (Boström et al., 2021) fit a separate residual law per region, and Toccaceli (2026) chooses the bins by minimizing a leave-one-out CRPS. These produce sharp, shape-adaptive predictive distributions, and they do so by leaving the single-shape class of Proposition 2 (a per-bin shape is X -dependent) and, in the CRPS case, by making the partition criterion a proper score. This is the most pointed case for the defence, and the concession is real: when the reference class is itself selected by a proper scoring rule, the conformal *construction* is performing conditional density estimation. The category distinction does not vanish; it relocates to “which component estimates the distribution,” and the answer is the conditional partition, not the coverage certificate: the value is in the estimation, as throughout. Conformalized quantile regression (Romano et al., 2019) is the same story with a quantile model in place of the bins.

End-to-end conformal training. The orthogonality of Proposition 1 is a statement about *post-hoc* conformalization of a fixed predictor. Conformal training (Stutz et al., 2022; Correia et al., 2024) deliberately differentiates through the conformalizer to minimize expected set size subject to coverage, which couples the coverage and efficiency axes by construction. This does not contradict the proposition; it illustrates its corollary: to obtain sharpness you must optimize for it, rather than read it off the coverage certificate.

The information in set size. Correia et al. (2024) connect conformal set size to information-theoretic uncertainty, with bounds relating expected prediction-set size to the conditional entropy $H(Y | X)$. This is complementary to our Proposition 3: both say conformal *geometry* carries information. It also sharpens the two-coordinate plane of Section 7: the *efficiency* (set-size / sharpness) coordinate is the information-bearing one, while the marginal *coverage* coordinate, by Proposition 1, is not. Reporting only the latter discards exactly the coordinate that encodes aleatoric uncertainty.

Approaching conditional coverage. The no-go result of Section 4 concerns *exact, distribution-free, finite-length* conditional coverage. It is not the end of the subject. Diagnostics such as the excess-risk-of-target-coverage family (Braun et al., 2025) quantify and decompose the conditional gap; methods that target conditional or subgroup-wise validity directly interpolate between marginal and conditional coverage (Gibbs et al., 2025); and post-processing via pivotal scores and optimal transport (Laplante, 2026) attains *approximate* conditional coverage by estimating the conditional law of the score, again conditioning on X , and explicitly under the structural assumptions the impossibility theorem rules out for the exact, assumption-free case. These approach the goalpost; they do not move the one the no-go result defends.

Modern time-series conformal. Beyond the adaptive methods benchmarked in Section 8, methods such as HopCPT (Auer et al., 2023) exploit temporal structure to tighten intervals and improve empirical adaptivity. Their guarantees remain of the approximate / long-run kind catalogued there;

the improvement, again, comes from modeling the conditional error law.

Beyond the log-score. We used the log-score for the clean information-theoretic identities. The two-coordinate distinction (coverage versus score) carries over to any strictly proper score, but the exact mutual-information identity is special to the log-score; for CRPS or the interval score the analogous gap is the regret of the best X -blind residual law under that score. The accompanying benchmark in fact ranks methods by the interval (Winkler) score and CRPS (Gneiting and Raftery, 2007), on which the same orderings hold.

11 Prediction versus verification

Conformal prediction *verifies* a coverage property. It does not *model*, and a fixed rule cannot substitute for modeling. Conformalization is best viewed as a terminal certification operator: it repairs the marginal coverage coordinate of a set-valued report, but any improvement in a proper score must come from upstream modeling of the conditional location, scale, quantiles, or residual shape. It moves only the coverage coordinate; the residual-information gap is reduced only by conditioning on X (the conditional-shape models of Section 10), never by the conformal step.

Practical remarks. (i) Conformalize *last*: model first, then add the certificate, since the conformal step cannot improve any proper score (estimate first, certify second). (ii) Use a proper score (log-likelihood, CRPS) as the sharpness diagnostic; because conformalization leaves it unchanged, a score that does not move after conformalizing confirms that any gain came from modeling. (iii) To close $I(R; X)$, add conditioning to the model (heteroscedastic spread, quantile or residual-shape estimation), not tightness to the conformal step.

Limitations. The exact result is deliberately narrow, and we state where it stops. (i) *Scope.* The strong claims concern post-hoc, marginal, split conformal with a residual score and the single-shape residual class; the conditional constructions of Section 10 (CQR, Mondrian/binning CPS, conformal training) leave that class by design and are out of scope as targets of the critique. (ii) *Fixed location.* The gap $I(R; X)$ is defined for a fixed $\hat{\mu}$ and measures conditional residual shape, not the quality of the point forecast (Remark 4). (iii) *Score-specific identity.* The exact mutual-information form is special to the log-score; for CRPS or the interval score the analogous gap is the regret of the best X -blind residual law under that score, not an information quantity. (iv) *Asymptotic density statement.* The density/log-score equalities of Propositions 2–3 are a large-calibration (or smoothed-density) statement; the object conformal delivers finite-sample is coverage, not a density. (v) *Synthetic illustrations.* The figures and benchmark use synthetic generators with known ground truth, which is what makes the oracle and $I(R; X)$ computable; the identity itself is verified numerically by `check_gap.py`.

12 Conclusion

Conformal prediction certifies the coverage of a set, finite-sample and distribution-free, under exchangeability. That guarantee is genuine and, for the right objective, exact. The error is reading it as distributional forecast quality. For a fixed location predictor, the guarantee you can have (marginal) is rarely the one you want (conditional), and the conditional one is not available distribution-free without more structure. The output is a set, not a measure. Read as a single-shape distribution, its log-score loss against the oracle is exactly the residual-information term $I(R; X)$ of (4). And the exchangeability even the basic guarantee needs is usually missing in deployed

time-series. Within the single-shape residual class it is optimal, and where a guarantee is the product it is the right tool. But coverage answers a question, *is the truth in this region?*, that, in forecasting, is often not the one you were asking. Conformal prediction is not weak uncertainty quantification. It is exact uncertainty quantification for a set — a coverage guarantee, no more and no less. The category error is reading that certificate as a verdict on the forecast underneath it.

Reproducibility. Every data figure is generated by the accompanying script `figures.py` (Figures 1–3, 4, 5, 7), and the time-series illustration by the benchmark scripts, all from the synthetic generators described in the text; Figure 6 is a schematic drawn by `fig_plane_schematic.py`. The identities of Proposition 3 are confirmed numerically by `check_gap.py`, which computes both sides independently (by grid integration) and matches them to machine precision on a heteroscedastic Gaussian and a skewed residual law. Code, figures, and an interactive companion are available at <https://conformalprediction.net> and <https://github.com/microprediction/conformalprediction>.

Data Availability Statement. This study uses no external or third-party data. All figures and numerical results are produced from the synthetic data-generating processes described in the text; the code that implements them and reproduces every figure, together with `check_gap.py` (the machine-precision numerical verification of Proposition 3), is openly available at <https://github.com/microprediction/conformalprediction>.

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